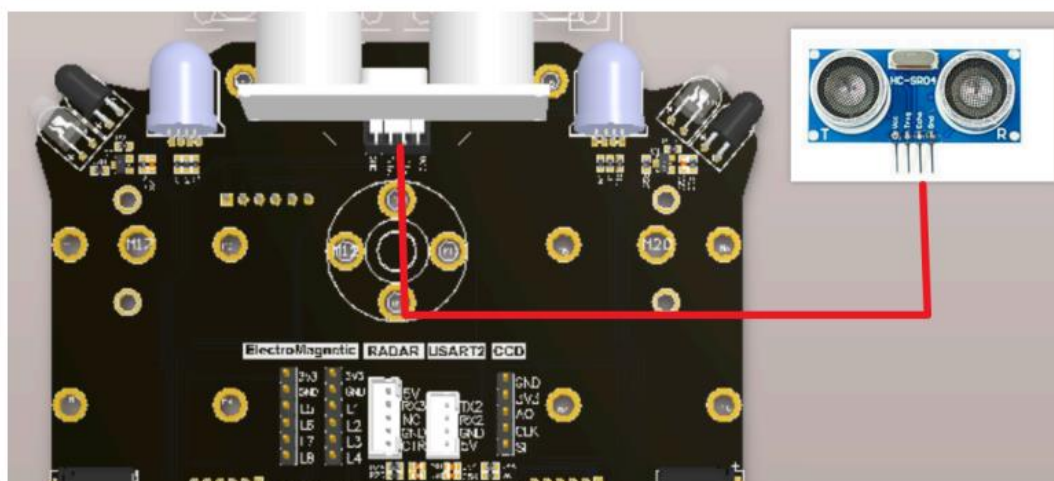
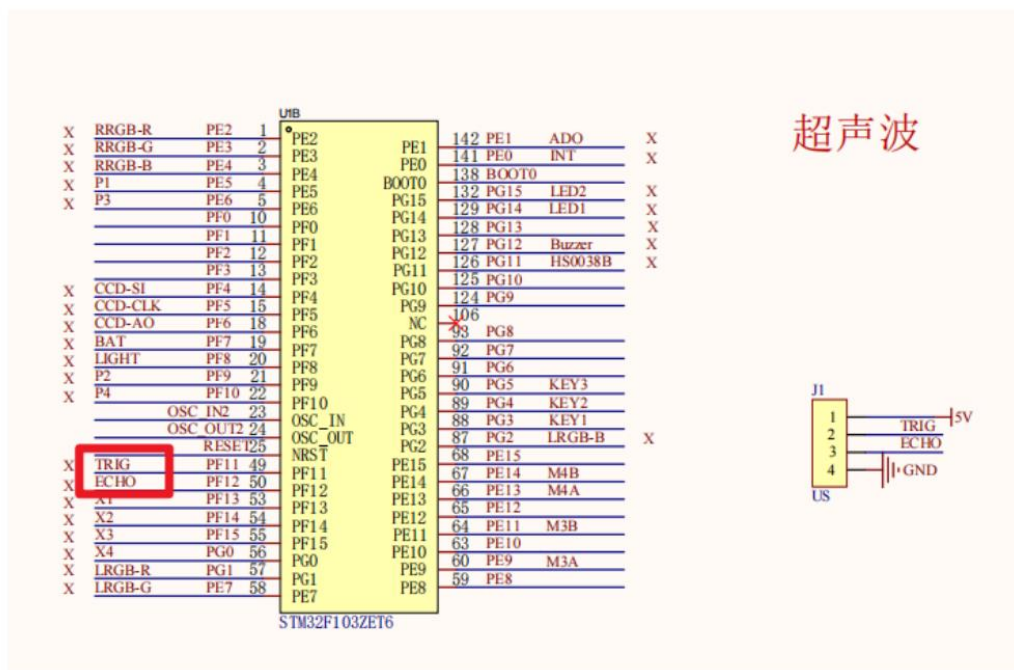


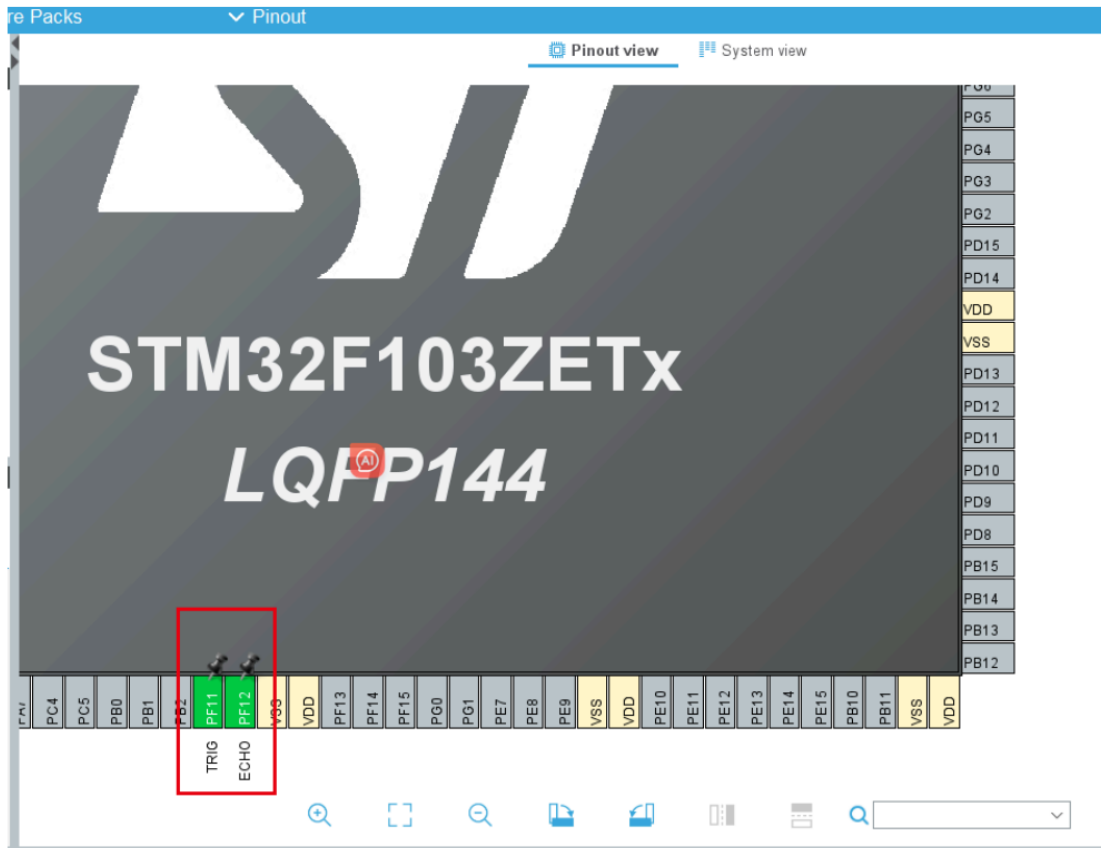
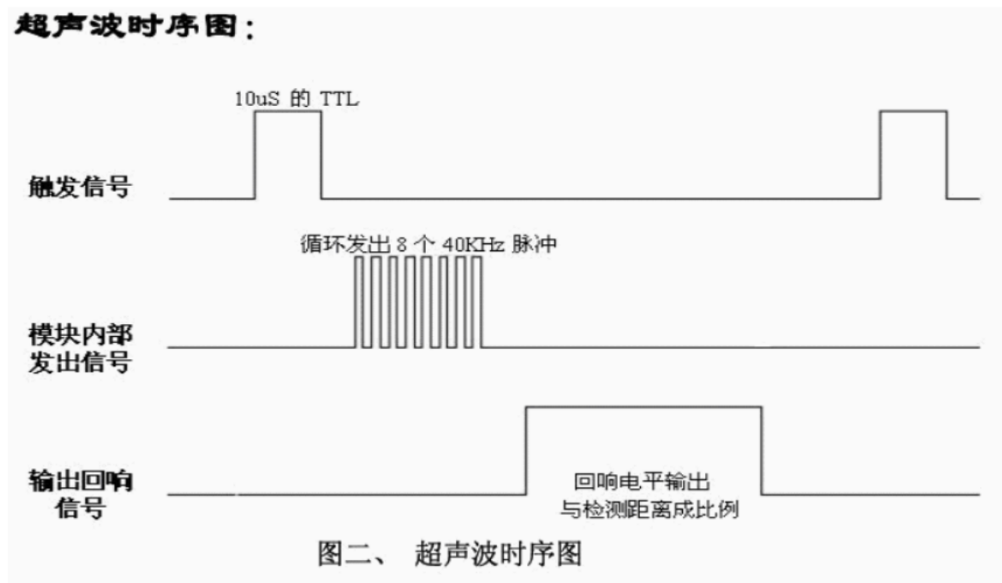


(原理图名称)	控制引脚	具体含义
TRIG	PF11	触发端
ECHO	PF12	接收端

本实验超声波模块信息:

型号	HC-SR04	探测距离	2-400cm
工作电压	5V	高精度	可达0.3cm
工作电流	15mA	盲区	2cm
工作频率	40KHz	引脚线序	VCC、Trig (控制端)、Echo(接收端)、GND
静态工作电流	<2mA	输入触发信号	10us的TTL脉冲
感应角度	不大于15°	输入回响信号	输出TTL电平信号，与射程成比例
射程范围	2cm-4m (峰值)	电平输出	TTL电平

测距原理：在超声波模块的触发端输入 10 微秒以上的高电位，即可发射超声波，发射超声波之后，与接收到传回的超声波之前，接收端是高电位。因此，程序可以从“响应”脚位的高位脉冲持续时间，换算出被测物的距离。测试距离=(高电平时间*声速(340M/S))/2;



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GPIO Mode and Configuration

Configuration

Group By Peripherals

✓ GPIO

✓ RCC

✓ SYS

Search Signals

Search (Ctrl+F)

☐ Show only Modified Pins

Pin Na...	Signal on	GPIO outp...	GPIO mode	GPIO Pull...	Maximum...	User Label	Modified
PF11	n/a	High	Output Pu...	No pull-up...	High	TRIG	✓
PF12	n/a	n/a	Input mode	No pull-up...	n/a	ECHO	✓

PF11 Configuration :

GPIO output level

High

GPIO mode

Output Push Pull

GPIO Pull-up/Pull-down

No pull-up and no pull-down

Maximum output speed

High

User Label

TRIG

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NVIC

Code generation

Priority Group

..

☐ Sort by Preemption Priority and Sub Priority

☐ Sort by interrupts names

Search

ⓘ

 Show

available interrupts

☒ Force DMA channels Interru

NVIC Interrupt Table

Enabled

Preemption Priority

Sub Prior

Non maskable interrupt	☒	0	0
Hard fault interrupt	☒	0	0
Memory management fault	☒	0	0
Prefetch fault, memory access fault	☒	0	0
Undefined instruction or illegal state	☒	0	0
System service call via SWI instruction	☒	0	0
Debug monitor	☒	0	0
Pendable request for system service	☒	0	0
Time base: System tick timer	☒	2	0
PVD interrupt through EXTI line 16	☐	0	0
Flash global interrupt	☐	0	0
RCC global interrupt	☐	0	0
USART1 global interrupt	☐	0	0
TIM7 global interrupt	☒	1	1

☐ Enabled

Preemption Priority

⌵

Sub Priority

⌵

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NVIC Mode and Configuration

Configuration

NVIC

Code generation

Enabled interrupt table	Select for init sequen...	Generate Enable...	Generat...	Call HAL h...
Non maskable interrupt	☐	☐	☒	☐
Hard fault interrupt	☐	☐	☒	☐
Memory management fault	☐	☐	☒	☐
Prefetch fault, memory acc...	☐	☐	☒	☐
Undefined instruction or ille...	☐	☐	☒	☐
System service call via SW...	☐	☐	☒	☐
Debug monitor	☐	☐	☒	☐
Pendable request for syste...	☐	☐	☒	☐
Time base: System tick ti...	☐	☐	☒	☒
TIM7 global interrupt	☐	☒	☒	☒

Interrupt unmasking ordering table (interrupt init code is moved after all the peripheral init code)

Rank	Interrupt name
------	----------------

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TIM7 Mode and Configuration

Mode

☒ Activated

☐ One Pulse Mode

Configuration

Reset Configuration

☒ Parameter Settings

☒ User Constants

☒ NVIC Settings

☒ DMA Settings

Configure the below parameters :

Search (Ctrl+F)

☒ Counter Settings

Prescaler (PSC - 16 bits value)71

Counter ModeUp

Counter Period (AutoReload Register -...9

auto-reload preloadDisable

☒ Trigger Output (TRGO) Parameters

Trigger Event SelectionReset (UG bit from TIMx_EGR)

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TIM7 Mode and Configuration

Mode

☒ Activated

☐ One Pulse Mode

Configuration

Reset Configuration

✓ Parameter Settings

✓ User Constants

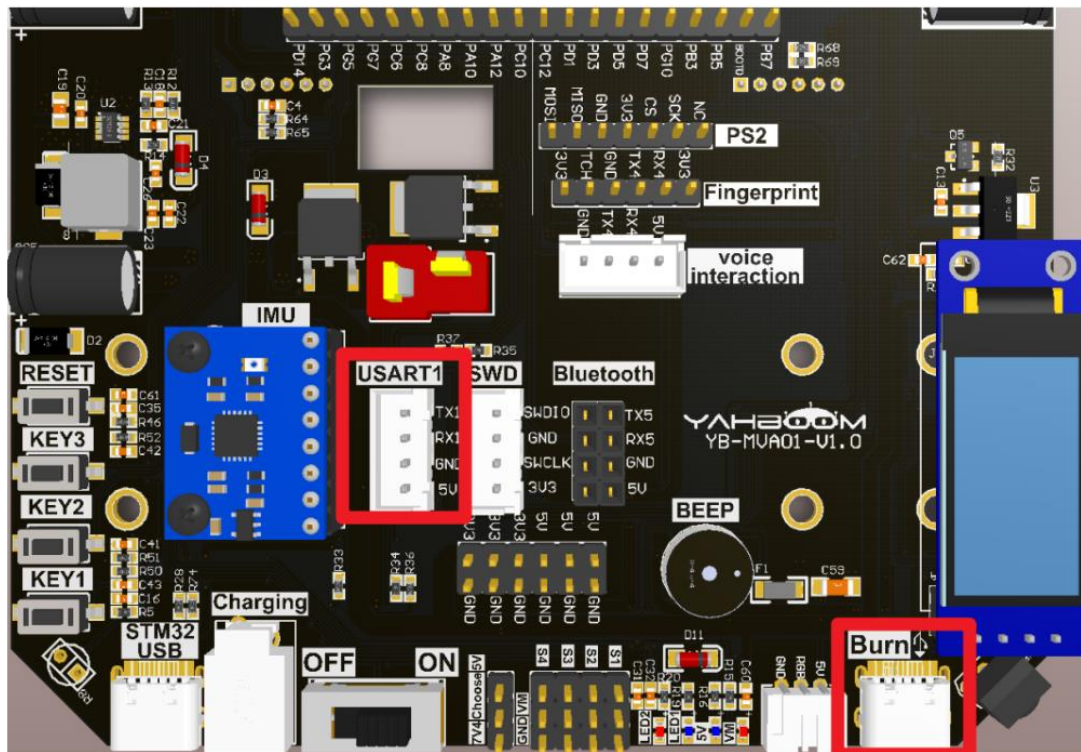
✓

 NVIC Settings

✓ DMA Settings

NVIC Interrupt Table	Enabled	Preemption Priority	Sub Priority
TIM7 global interrupt	☒	1	1

USART 1 阻塞发送串口配置



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▼ Software

USART1 Mode and Configuration

Mode

Mode

Asynchronous

Hardware Flow Control (RS232)

Disable

Configuration

Reset Configuration

✓ NVIC Settings

✓ DMA Settings

✓ GPIO Settings

✓ Parameter Settings

✓ User Constants

Configure the below parameters :

Search (Ctrl+F)

Basic Parameters

Baud Rate

Word Length

Parity

Stop Bits

Advanced Parameters

Data Direction

Over Sampling

115200 Bits/s

8 Bits (including Parity)

None

1

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Mode

Mode

Asynchronous

Hardware Flow Control (RS232)

Disable

Configuration

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✓ NVIC Settings

✓ DMA Settings

✓ GPIO Settings

✓ Parameter Settings

✓ User Constants

Configure the below parameters :

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Basic Parameters

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Word Length

Parity

Stop Bits

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Over Sampling

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8 Bits (including Parity)

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1

Receive and Transmit

16 Samples